A8

Q1

a): Prove f is not Lipschitz smooth

Since $f(x) = |x|^{3/2}$ involves an absolute value, handle $x > 0$, $x < 0$, and $x = 0$ separately.

- **For $x > 0$:**
  $f(x) = x^{3/2}$, so

$$f'(x) = \frac{d}{dx}(x^{3/2}) = \frac{3}{2}x^{1/2} = \frac{3}{2}x^{1/2}, \quad x > 0.$$

- **For $x < 0$:**
  Here, $|x| = -x$, and since $x < 0$, $-x > 0$, so $f(x) = (-x)^{3/2}$. Using the chain rule, let $u = -x$, where $u > 0$, and $f(x) = u^{3/2}$. Then:

$$\frac{df}{dx} = \frac{df}{du} \cdot \frac{du}{dx}, \quad \frac{df}{du} = \frac{3}{2}u^{1/2}, \quad \frac{du}{dx} = -1,$$

$$f'(x) = \frac{3}{2}u^{1/2} \cdot (-1) = -\frac{3}{2}(-x)^{1/2}, \quad x < 0.$$

- **For $x = 0$:**
  Check differentiability using the limit definition:

$$f'(0) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \to 0} \frac{|h|^{3/2}}{h}.$$

  - Right-hand limit ($h \to 0^+$):

$$\frac{h^{3/2}}{h} = h^{1/2} \to 0.$$

  - Left-hand limit ($h \to 0^-$):

$$\frac{|h|^{3/2}}{h} = \frac{(-h)^{3/2}}{h} = (-h)^{1/2} \cdot (-h)/h = -(-h)^{1/2} \to 0.$$

    Since both limits equal 0, $f'(0) = 0$.

Thus, the derivative is:

$$f'(x) = \begin{cases} \frac{3}{2}x^{1/2} & \text{if } x > 0, \\ 0 & \text{if } x = 0, \\ -\frac{3}{2}(-x)^{1/2} & \text{if } x < 0. \end{cases}$$

Alternatively:

$$f'(x) = \frac{3}{2}|x|^{1/2}\text{sign}(x),$$

To show $f'(x)$ is not Lipschitz continuous, compute $\frac{|f'(x)-f'(y)|}{|x-y|}$ and see if it's bounded. Consider $y = 0$ (where $f'(0) = 0$) and $x > 0$:

$$|f'(x) - f'(0)| = \left|\frac{3}{2}x^{1/2} - 0\right| = \frac{3}{2}x^{1/2},$$

$$|x - 0| = x,$$

$$\frac{|f'(x) - f'(0)|}{|x - 0|} = \frac{\frac{3}{2}x^{1/2}}{x} = \frac{3}{2}x^{-1/2}.$$

As $x \to 0^+$, $x^{-1/2} = \frac{1}{\sqrt{x}} \to \infty$. For any $L > 0$, choose x small enough, e.g., $x = \left(\frac{3}{2(L+1)}\right)^2$:

$$x^{-1/2} = \left(\left(\frac{3}{2(L+1)}\right)^2\right)^{-1/2} = \frac{2(L+1)}{3},$$

$$\frac{3}{2}x^{-1/2} = \frac{3}{2} \cdot \frac{2(L+1)}{3} = L + 1 > L.$$

Thus, $|f'(x) - f'(0)| > L|x - 0|$, violating the condition for any fixed L when x is sufficiently small. Hence, no L satisfies the inequality for all x, y, and $f'(x)$ is not Lipschitz continuous.

b): Gradient descent step size

Choose $x_0 > 0$ small, since the derivative behaves sharply near 0. Then:

$$f'(x_0) = \frac{3}{2}x_0^{1/2} > 0,$$

$$x_1 = x_0 - \bar{\alpha} f'(x_0) = x_0 - \bar{\alpha} \cdot \frac{3}{2} x_0^{1/2}.$$

Since $\bar{\alpha} > 0$ and $f'(x_0) > 0$, $x_1 < x_0$. We want $f(x_1) = |x_1|^{3/2} \geq f(x_0) = x_0^{3/2}$. Given $f(x)$ is even and increasing in $|x|$, if $x_1 < 0$ and $|x_1| > x_0$, then $f(x_1) = |x_1|^{3/2} > x_0^{3/2} = f(x_0)$.

- $x_1 < 0$**:**

$$x_0 - \bar{\alpha} \cdot \frac{3}{2} x_0^{1/2} < 0 \implies \bar{\alpha} \cdot \frac{3}{2} x_0^{1/2} > x_0 \implies \bar{\alpha} > \frac{2}{3} x_0^{1/2}.$$

  Since $\bar{\alpha}$ is fixed, for small $x_0 > 0$, $\frac{2}{3} x_0^{1/2} \to 0$, so this holds.
- $|x_1| > x_0$**:**
  If $x_1 < 0$, then $|x_1| = -x_1 = \bar{\alpha} \cdot \frac{3}{2} x_0^{1/2} - x_0$. We need:

$$|x_1| > x_0 \implies \bar{\alpha} \cdot \frac{3}{2} x_0^{1/2} - x_0 > x_0 \implies \bar{\alpha} \cdot \frac{3}{2} x_0^{1/2} > 2x_0 \implies \bar{\alpha} > \frac{4}{3} x_0^{1/2}.$$

  As $x_0 \to 0^+$, $\frac{4}{3} x_0^{1/2} \to 0$, and since $\bar{\alpha} > 0$ is fixed, this is true for sufficiently small x_0.

For small $x_0 > 0$ where $\bar{\alpha} > \frac{4}{3} x_0^{1/2}$:

- $x_1 < 0$ (from above),
- $|x_1| = \bar{\alpha} \cdot \frac{3}{2} x_0^{1/2} - x_0 > x_0$,
- $f(x_1) = |x_1|^{3/2} > x_0^{3/2} = f(x_0)$, since $|x_1| > x_0 > 0$ and $t \mapsto t^{3/2}$ is increasing for $t > 0$.

Thus, $f(x_1) > f(x_0)$, which satisfies $f(x_1) \geq f(x_0)$.

Q2

a): Show that the function is not convex and verify $x^* = 0$ is a global minimizer

$$f'(x) = 2x + 3 \sin 2x.$$
$$f''(x) = 2 + 6 \cos 2x.$$

- **Convexity check**:
  Since $\cos 2x$ ranges from -1 to 1:
  - When $\cos 2x = 1$, $f''(x) = 2 + 6 \cdot 1 = 8 > 0$.
  - When $\cos 2x = -1$, $f''(x) = 2 + 6 \cdot (-1) = 2 - 6 = -4 < 0$.
    Because $f''(x)$ can be negative (e.g., at $x = \frac{\pi}{2}$, where $2x = \pi$, $\cos \pi = -1$), the function is not convex.
- **Global minimizer at $x^* = 0$**:
  - Evaluate $f(0) = 0^2 + 3 \sin^2 0 = 0$.
  - For any x, $f(x) = x^2 + 3 \sin^2 x$, where $x^2 \geq 0$ and $3 \sin^2 x \geq 0$. Thus, $f(x) \geq 0 = f(0)$.
  - Equality holds when $x^2 = 0$ and $\sin^2 x = 0$, i.e., $x = 0$ (at $x = n\pi$, $n \neq 0$, $\sin x = 0$ but $x^2 > 0$, so $f(x) > 0$).
  - Check the critical point: $f'(0) = 2 \cdot 0 + 3 \sin 0 = 0$, and $f''(0) = 2 + 6 \cos 0 = 8 > 0$, confirming a local minimum. Since $f(x) \geq f(0)$ everywhere, $x^* = 0$ is a global minimizer.

b): Show that the function is Lipschitz smooth with some constant $L > 0$

- From part (a), $f''(x) = 2 + 6 \cos 2x$.
- Since $|\cos 2x| \leq 1$,

$$|f''(x)| = |2 + 6 \cos 2x| \leq 2 + 6|\cos 2x| \leq 2 + 6 \cdot 1 = 8.$$

- The maximum occurs when $|\cos 2x| = 1$:
  - $\cos 2x = 1$, $f''(x) = 8$,
  - $\cos 2x = -1$, $f''(x) = -4$, so $|f''(x)| \leq 8$.
- Thus, $L = 8$ is a valid constant, as $|f''(x)| \leq 8$ for all x, implying $f'(x)$ is Lipschitz continuous with constant $L = 8$.

Hence, f is Lipschitz smooth with $L = 8$.

c): Gradient descent inequality

Use Lemma 1: if g is Lipschitz smooth with constant L, then

$$g(y) \leq g(x) + g'(x)(y - x) + \frac{L}{2}(y - x)^2.$$

- Gradient descent update: $x_{k+1} = x_k - \bar{\alpha} f'(x_k)$.
- Apply Lemma 1 to f, with $x = x_k$, $y = x_{k+1}$:

$$f(x_{k+1}) \leq f(x_k) + f'(x_k)(x_{k+1} - x_k) + \frac{L}{2}(x_{k+1} - x_k)^2.$$

- Substitute $x_{k+1} - x_k = -\bar{\alpha} f'(x_k)$:

$$f(x_{k+1}) \leq f(x_k) + f'(x_k)(-\bar{\alpha} f'(x_k)) + \frac{L}{2}(-\bar{\alpha} f'(x_k))^2.$$

- Simplify:
  - $f'(x_k)(-\bar{\alpha} f'(x_k)) = -\bar{\alpha}[f'(x_k)]^2$,
  - $(-\bar{\alpha} f'(x_k))^2 = \bar{\alpha}^2[f'(x_k)]^2$,
    so

$$f(x_{k+1}) \leq f(x_k) - \bar{\alpha}[f'(x_k)]^2 + \frac{L}{2}\bar{\alpha}^2[f'(x_k)]^2.$$

- Factorize:

$$f(x_{k+1}) \leq f(x_k) + \left(-\bar{\alpha} + \frac{L\bar{\alpha}^2}{2}\right)[f'(x_k)]^2 = f(x_k) - \left(\bar{\alpha} - \frac{L\bar{\alpha}^2}{2}\right)[f'(x_k)]^2.$$

- Since $[f'(x_k)]^2 = |f'(x_k)|^2$, the inequality holds as required.

d): Linear convergence

- From part (c):

$$f(x_{k+1}) \leq f(x_k) - \left(\bar{\alpha} - \frac{L\bar{\alpha}^2}{2}\right)|f'(x_k)|^2.$$

- Subtract $f(x^*)$ (where $f(x^*) = f(0) = 0$):

$$f(x_{k+1}) - f(x^*) \leq f(x_k) - f(x^*) - \left(\bar{\alpha} - \frac{L\bar{\alpha}^2}{2}\right)|f'(x_k)|^2.$$

- Substitute the given condition:

$$|f'(x_k)|^2 \geq \frac{1}{16}(f(x_k) - f(x^*)),$$

  so

$$f(x_{k+1}) - f(x^*) \leq f(x_k) - f(x^*) - \left(\bar{\alpha} - \frac{L\bar{\alpha}^2}{2}\right) \cdot \frac{1}{16}(f(x_k) - f(x^*)).$$

- Factor out $f(x_k) - f(x^*)$:

$$f(x_{k+1}) - f(x^*) \leq \left[1 - \frac{1}{16}\left(\bar{\alpha} - \frac{L\bar{\alpha}^2}{2}\right)\right](f(x_k) - f(x^*)).$$

- Define $\eta = 1 - \frac{1}{16}\left(\bar{\alpha} - \frac{L\bar{\alpha}^2}{2}\right)$. We need $\eta \in (0, 1)$, i.e.,

$$0 < \frac{1}{16}\left(\bar{\alpha} - \frac{L\bar{\alpha}^2}{2}\right) < 1.$$

- Let $t = \bar{\alpha}$, then $h(t) = t - \frac{Lt^2}{2}$ must be positive and $\frac{h(t)}{16} < 1$.
  - $h(t) = t(1 - \frac{Lt}{2})$, positive when $0 < t < \frac{2}{L}$.
  - Maximize $h(t)$: $h'(t) = 1 - Lt = 0$, so $t = \frac{1}{L}$, and $h\left(\frac{1}{L}\right) = \frac{1}{L} - \frac{L}{2} \cdot \frac{1}{L^2} = \frac{1}{2L}$.
  - Then, $\frac{1}{16} \cdot \frac{1}{2L} = \frac{1}{32L}$.
- With $L = 8$, $\eta = 1 - \frac{1}{32 \cdot 8} \approx 0.996$, but we can use any $L \geq 8$. For generality, choose $\bar{\alpha} = \frac{1}{L}$:

$$\eta = 1 - \frac{1}{32L} < 1,$$

  and since $L > 0$, $\eta > 0$, so $\eta \in (0, 1)$.

Thus, with $\bar{\alpha} = \frac{1}{L}$, linear convergence is achieved, leveraging the given condition despite non-convexity, aligning with the remark.

Q3

Python Implementation

```python
import math

# Define the objective function
def f(x):
    return 4 * x**2 - x**4

# Define the derivative of the function
def df(x):
    return 8 * x - 4 * x**3

# Bisection Method
```

```
def bisection_method(a, b, tol):
    iterations = 0
    while b - a >= tol:
        m = (a + b) / 2
        derivative = df(m)
        if derivative > 0:
            b = m
        elif derivative < 0:
            a = m
        else:
            break  # Exact minimum found (unlikely with floating-point)
        iterations += 1
    return (a + b) / 2, iterations

# Golden Section Method
def golden_section_method(a, b, tol):
    rho = (math.sqrt(5) - 1) / 2  # Golden ratio conjugate ≈ 0.618
    iterations = 0
    while b - a >= tol:
        x1 = a + (1 - rho) * (b - a)
        x2 = a + rho * (b - a)
        if f(x1) > f(x2):
            a = x1
        else:
            b = x2
        iterations += 1
    return (a + b) / 2, iterations

# Parameters
a_initial = -1.0
b_initial = 0.5
tolerance = 2e-4  # Interval length to ensure midpoint is within 1e-4 of x*

# Run Bisection Method
approx_bisection, iter_bisection = bisection_method(a_initial, b_initial, tolerance)
print("Bisection Method:")
print(f"Approximate solution: {approx_bisection}")
print(f"Number of iterations: {iter_bisection}")

# Run Golden Section Method
approx_golden, iter_golden = golden_section_method(a_initial, b_initial, tolerance)
print("Golden Section Method:")
print(f"Approximate solution: {approx_golden}")
print(f"Number of iterations: {iter_golden}")

# Comparison
print("\nComparison:")
print(f"Bisection Method required {iter_bisection} iterations.")
print(f"Golden Section Method required {iter_golden} iterations.")
```

Results

```
Bisection Method:
Approximate solution: 3.0517578125e-05
Number of iterations: 13

Golden Section Method:
Approximate solution: -1.4115521418356513e-05
Number of iterations: 19

Comparison:
Bisection Method required 13 iterations.
Golden Section Method required 19 iterations.
```

Q4

a): True

A direction d is a descent direction at $\boldsymbol{x}_k$ if the directional derivative in that direction is negative, i.e., $\nabla f(\boldsymbol{x}_k)^T d < 0$. Here, the direction is v, defined component-wise based on the gradient $\nabla f(\boldsymbol{x}_k)$. Let's compute the inner product $\nabla f(\boldsymbol{x}_k)^T v = \sum_i \nabla f(\boldsymbol{x}_k)_i v_i$ by considering each case:

- **Case 1:** If $\nabla f(\boldsymbol{x}_k)_i > 0$, then $v_i = -1$, so $\nabla f(\boldsymbol{x}_k)_i v_i = \nabla f(\boldsymbol{x}_k)_i \cdot (-1) = -\nabla f(\boldsymbol{x}_k)_i < 0$.
- **Case 2:** If $\nabla f(\boldsymbol{x}_k)_i < 0$, then $v_i = 1$, so $\nabla f(\boldsymbol{x}_k)_i v_i = \nabla f(\boldsymbol{x}_k)_i \cdot 1 = \nabla f(\boldsymbol{x}_k)_i < 0$.
- **Case 3:** If $\nabla f(\boldsymbol{x}_k)_i = 0$, then $v_i = 0$, so $\nabla f(\boldsymbol{x}_k)_i v_i = 0 \cdot 0 = 0$.

Each term in the sum is either negative (when the gradient component is nonzero) or zero (when it is zero). Since $\boldsymbol{x}_k$ is not a stationary point, $\nabla f(\boldsymbol{x}_k) \neq 0$, meaning at least one component of the gradient is nonzero. Thus, the sum includes at least one negative term and no positive terms, ensuring $\nabla f(\boldsymbol{x}_k)^T v < 0$. Hence, v is a descent direction

under this condition.

Justification: The direction v is constructed to oppose the gradient's sign in each component, making $\nabla f(\boldsymbol{x}_k)^T v < 0$ whenever $\nabla f(\boldsymbol{x}_k) \neq 0$, satisfying the definition of a descent direction.

---

b): True

In gradient descent, the search direction is $d = -\nabla f(\boldsymbol{x}_k)$. First, verify that this is a descent direction:

$$\nabla f(\boldsymbol{x}_k)^T d = \nabla f(\boldsymbol{x}_k)^T (-\nabla f(\boldsymbol{x}_k)) = -\|\nabla f(\boldsymbol{x}_k)\|^2.$$

Since $\boldsymbol{x}_k$ is not a stationary point, $\nabla f(\boldsymbol{x}_k) \neq 0$, so $\nabla f(\boldsymbol{x}_k)^T d < 0$, confirming d is a descent direction.

Backtracking line search selects a step size $\alpha_k > 0$ satisfying the Armijo condition:

$$f(\boldsymbol{x}_k + \alpha_k d) \leq f(\boldsymbol{x}_k) + c\alpha_k \nabla f(\boldsymbol{x}_k)^T d,$$

where $c \in (0, 1)$ (e.g., $c = 10^{-4}$). Substituting $d = -\nabla f(\boldsymbol{x}_k)$, we get:

$$\nabla f(\boldsymbol{x}_k)^T d = -\|\nabla f(\boldsymbol{x}_k)\|^2 < 0,$$

so the right-hand side becomes:

$$f(\boldsymbol{x}_k) + c\alpha_k(-\|\nabla f(\boldsymbol{x}_k)\|^2) = f(\boldsymbol{x}_k) - c\alpha_k \|\nabla f(\boldsymbol{x}_k)\|^2,$$

which is less than $f(\boldsymbol{x}_k)$ since $c\alpha_k \|\nabla f(\boldsymbol{x}_k)\|^2 > 0$. Backtracking starts with an initial α (e.g., 1) and reduces it by a factor $\rho \in (0, 1)$ until the condition holds. Since f is differentiable and d is a descent direction, Taylor's expansion gives:

$$f(\boldsymbol{x}_k + \alpha d) = f(\boldsymbol{x}_k) + \alpha \nabla f(\boldsymbol{x}_k)^T d + o(\alpha),$$

and because $\nabla f(\boldsymbol{x}_k)^T d < 0$ and $c < 1$, for sufficiently small α, the decrease exceeds the linear term scaled by c, ensuring the inequality is strict: $f(\boldsymbol{x}_k + \alpha_k d) < f(\boldsymbol{x}_k)$. Thus, $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k d$ satisfies $f(\boldsymbol{x}_{k+1}) < f(\boldsymbol{x}_k)$.

Justification: The descent direction and backtracking line search guarantee a step size $\alpha_k > 0$ that reduces the function value when $\nabla f(\boldsymbol{x}_k) \neq 0$.

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c): True

The direction is again $d = -\nabla f(\boldsymbol{x}_k)$, a descent direction since $\nabla f(\boldsymbol{x}_k)^T d = -\|\nabla f(\boldsymbol{x}_k)\|^2 < 0$ when $\boldsymbol{x}_k$ is not stationary. Exact line search chooses $\alpha_k = \arg\min_{\alpha \geq 0} f(\boldsymbol{x}_k + \alpha d)$. Since d is a descent direction, consider the function along the line:

$$\phi(\alpha) = f(\boldsymbol{x}_k + \alpha d).$$

At $\alpha = 0$, $\phi(0) = f(\boldsymbol{x}_k)$, and the derivative is:

$$\phi'(0) = \nabla f(\boldsymbol{x}_k)^T d = -\|\nabla f(\boldsymbol{x}_k)\|^2 < 0,$$

so $\phi(\alpha) < \phi(0)$ for small $\alpha > 0$. Exact line search finds the global minimum of $\phi(\alpha)$. If f is bounded below (a common assumption in optimization), $\phi(\alpha)$ has a minimizer $\alpha_k \geq 0$. Since $\phi(\alpha)$ decreases initially, the minimum value satisfies $\phi(\alpha_k) \leq \phi(0)$, and because the derivative is negative at zero, $\alpha_k = 0$ cannot be the minimum (it's not a critical point of ϕ unless $d = 0$), so $\alpha_k > 0$ and $f(\boldsymbol{x}_{k+1}) = \phi(\alpha_k) < \phi(0) = f(\boldsymbol{x}_k)$.

Justification: Exact line search minimizes f along a descent direction, ensuring a decrease in function value when $\nabla f(\boldsymbol{x}_k) \neq 0$.

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d): False

Gradient descent with exact line search iteratively follows $d = -\nabla f(\boldsymbol{x}_k)$ with α_k minimizing f along that direction. While it decreases f each step (per part c) and often converges to a stationary point ($\nabla f = 0$) under conditions like f being bounded below, "always" converging to a local minimizer is stronger. Stationary points include local minimizers, maximizers, and saddle points. For non-convex f, convergence to a saddle point is possible.

Consider $f(x, y) = x^2 - y^2$, with $\nabla f = (2x, -2y)$. At $(0, 0)$, $\nabla f = 0$, and the Hessian $\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$ has eigenvalues 2 and -2, indicating a saddle point. Starting at $(0, y_0)$, $y_0 \neq 0$, the gradient is $(0, -2y_0)$, $d = (0, 2y_0)$, and $f(0, y_0 + \alpha 2y_0) = -(y_0(1 + 2\alpha))^2 \to -\infty$ as $\alpha \to \infty$. However, this function is unbounded below, so consider a bounded case like $f(x, y) = x^2 - y^2 + \frac{1}{4}y^4$. Stationary points occur where $\nabla f = (2x, -2y + y^3) = 0$, i.e., $(0, 0)$, $(0, \sqrt{2})$, and $(0, -\sqrt{2})$. At $(0, 0)$, the Hessian is indefinite (saddle); at $(0, \pm\sqrt{2})$, it's positive definite (minimizers). Depending on the starting point, gradient descent may approach $(0, 0)$, not a minimizer. In general, for non-convex functions, convergence to saddle points can occur, violating the "always" claim.

Justification: Gradient descent with exact line search can converge to saddle points, not just local minimizers, in non-convex functions.

Q5

a): Manual Computations

Two Steps with Constant Step Size $\alpha_k = 0.1$

Given $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A}\mathbf{x}$ with $\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$, the gradient is:

$$\nabla f(\mathbf{x}) = 2\mathbf{A}\mathbf{x} = 2 \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4x_1 \\ 10x_2 \end{bmatrix}$$

Start at $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

- **Step 1:**

$$\nabla f(\mathbf{x}_0) = \begin{bmatrix} 4 \cdot 1 \\ 10 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$$

$$\mathbf{x}_1 = \mathbf{x}_0 - 0.1 \cdot \nabla f(\mathbf{x}_0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.1 \cdot \begin{bmatrix} 4 \\ 10 \end{bmatrix} = \begin{bmatrix} 1 - 0.4 \\ 1 - 1 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0 \end{bmatrix}$$

- **Step 2:**

$$\nabla f(\mathbf{x}_1) = \begin{bmatrix} 4 \cdot 0.6 \\ 10 \cdot 0 \end{bmatrix} = \begin{bmatrix} 2.4 \\ 0 \end{bmatrix}$$

$$\mathbf{x}_2 = \mathbf{x}_1 - 0.1 \cdot \nabla f(\mathbf{x}_1) = \begin{bmatrix} 0.6 \\ 0 \end{bmatrix} - 0.1 \cdot \begin{bmatrix} 2.4 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.6 - 0.24 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.36 \\ 0 \end{bmatrix}$$

Result: After two steps, $\mathbf{x}_2 = \begin{bmatrix} 0.36 \\ 0 \end{bmatrix}$.

One Step with Exact Line Search

For exact line search, choose α to minimize $f(\mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k))$.

From $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

$$\nabla f(\mathbf{x}_0) = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$$

Let $\mathbf{x}(\alpha) = \mathbf{x}_0 - \alpha \nabla f(\mathbf{x}_0) = \begin{bmatrix} 1 - 4\alpha \\ 1 - 10\alpha \end{bmatrix}$.

$$f(\mathbf{x}(\alpha)) = \begin{bmatrix} 1 - 4\alpha \\ 1 - 10\alpha \end{bmatrix}^\top \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 - 4\alpha \\ 1 - 10\alpha \end{bmatrix} = 2(1 - 4\alpha)^2 + 5(1 - 10\alpha)^2$$

To find the minimum, compute the derivative:

$$\frac{d}{d\alpha} f(\alpha) = 2 \cdot 2(1 - 4\alpha)(-4) + 5 \cdot 2(1 - 10\alpha)(-10) = 1064\alpha - 116$$

Set to zero:

$$1064\alpha - 116 = 0 \Rightarrow \alpha = \frac{116}{1064} = \frac{29}{266}$$

Update:

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{29}{266} \cdot \begin{bmatrix} 4 \\ 10 \end{bmatrix} = \begin{bmatrix} \frac{75}{133} \\ -\frac{12}{133} \end{bmatrix}$$

Result: After one step, $\mathbf{x}_1 = \begin{bmatrix} \frac{75}{133} \\ -\frac{12}{133} \end{bmatrix}$.

One Step with Backtracking Line Search

Use the Armijo condition with $\gamma = \frac{1}{2}$, $\sigma = 0.2$, starting $\alpha = 1$.

From $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

$$\nabla f(\mathbf{x}_0) = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$$

$$f(\mathbf{x}_0) = 2 \cdot 1^2 + 5 \cdot 1^2 = 7$$

$$\|\nabla f(\mathbf{x}_0)\|^2 = 4^2 + 10^2 = 116$$

- **Try $\alpha = 1$:**

$$\mathbf{x}_0 - \alpha \nabla f(\mathbf{x}_0) = \begin{bmatrix} 1 - 4 \\ 1 - 10 \end{bmatrix} = \begin{bmatrix} -3 \\ -9 \end{bmatrix}$$

$$f(\begin{bmatrix} -3 \\ -9 \end{bmatrix}) = 2 \cdot (-3)^2 + 5 \cdot (-9)^2 = 18 + 405 = 423$$

$$7 - \frac{1}{2} \cdot 1 \cdot 116 = 7 - 58 = -51$$

$$423 \leq -51 \text{ is false}$$

  Update $\alpha = 1 \cdot 0.2 = 0.2$.

- **Try $\alpha = 0.2$:**

$$\mathbf{x}_0 - 0.2 \cdot \begin{bmatrix} 4 \\ 10 \end{bmatrix} = \begin{bmatrix} 1 - 0.8 \\ 1 - 2 \end{bmatrix} = \begin{bmatrix} 0.2 \\ -1 \end{bmatrix}$$

$$f(\begin{bmatrix} 0.2 \\ -1 \end{bmatrix}) = 2 \cdot (0.2)^2 + 5 \cdot (-1)^2 = 0.08 + 5 = 5.08$$

$$7 - \frac{1}{2} \cdot 0.2 \cdot 116 = 7 - 11.6 = -4.6$$

$$5.08 \leq -4.6 \text{ is false}$$

  Update $\alpha = 0.2 \cdot 0.2 = 0.04$.

- **Try $\alpha = 0.04$:**

$$\mathbf{x}_0 - 0.04 \cdot \begin{bmatrix} 4 \\ 10 \end{bmatrix} = \begin{bmatrix} 1 - 0.16 \\ 1 - 0.4 \end{bmatrix} = \begin{bmatrix} 0.84 \\ 0.6 \end{bmatrix}$$

$$f(\begin{bmatrix} 0.84 \\ 0.6 \end{bmatrix}) = 2 \cdot (0.84)^2 + 5 \cdot (0.6)^2 = 1.4112 + 1.8 = 3.2112$$

$$7 - \frac{1}{2} \cdot 0.04 \cdot 116 = 7 - 2.32 = 4.68$$

$$3.2112 \leq 4.68 \text{ is true}$$

Result: After one step, $\alpha = 0.04$, $\mathbf{x}_1 = \begin{bmatrix} 0.84 \\ 0.6 \end{bmatrix}$.

b): Python Implementation

Below is the Python code implementing the three strategies for 10 iterations, starting from $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

```python
import numpy as np

# Define matrix A
A = np.array([[2, 0], [0, 5]])

# Objective function
def f(x):
    return np.dot(x.T, np.dot(A, x))

# Gradient
def grad_f(x):
    return 2 * np.dot(A, x)

# Constant step size
def gd_constant_alpha(x0, alpha, num_iterations):
    x = x0.copy()
    for _ in range(num_iterations):
        x = x - alpha * grad_f(x)
    return x, f(x)

# Exact line search
def gd_exact_line_search(x0, num_iterations):
    x = x0.copy()
    for _ in range(num_iterations):
        grad = grad_f(x)
        alpha = np.dot(grad, grad) / (2 * np.dot(grad, np.dot(A, grad)))
        x = x - alpha * grad
    return x, f(x)

# Backtracking line search
def backtracking_line_search(x, grad, gamma=0.5, sigma=0.2):
    alpha = 1.0
    f_x = f(x)
    grad_norm_sq = np.dot(grad, grad)
    while f(x - alpha * grad) > f_x - gamma * alpha * grad_norm_sq:
        alpha *= sigma
    return alpha

def gd_backtracking(x0, num_iterations, gamma=0.5, sigma=0.2):
```

```
    x = x0.copy()
    for _ in range(num_iterations):
        grad = grad_f(x)
        alpha = backtracking_line_search(x, grad, gamma, sigma)
        x = x - alpha * grad
    return x, f(x)

# Initial point
x0 = np.array([1.0, 1.0])

# Run for 10 iterations
x_const, f_const = gd_constant_alpha(x0, 0.1, 10)
x_exact, f_exact = gd_exact_line_search(x0, 10)
x_back, f_back = gd_backtracking(x0, 10)

# Results
print("Constant step size (alpha = 0.1):")
print(f"Last iterate: {x_const}")
print(f"Objective value: {f_const:.10f}\n")

print("Exact line search:")
print(f"Last iterate: {x_exact}")
print(f"Objective value: {f_exact:.10f}\n")

print("Backtracking line search:")
print(f"Last iterate: {x_back}")
print(f"Objective value: {f_back:.10f}")
```

Results

Running the code yields:

```
Constant step size (alpha = 0.1):
Last iterate: [0.00604662 0.        ]
Objective value: 0.0000731232

Exact line search:
Last iterate: [8.44236534e-06 8.44236534e-06]
Objective value: 0.0000000005

Backtracking line search:
Last iterate: [ 0.04164315 -0.0100777 ]
Objective value: 0.0039761036
```

The exact line search and backtracking methods converge to the minimum $\mathbf{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ (where $f = 0$) rapidly due to the quadratic nature and positive definiteness of $\mathbf{A}$, while the constant step size converges more slowly.